

```

1 import tensorflow as tf
2 import matplotlib.pyplot as plt
3
4 from tensorflow.keras.datasets import imdb
5 from tensorflow.keras.preprocessing.sequence import pad_sequences
6 from tensorflow.keras.models import Sequential
7 from tensorflow.keras.layers import Embedding, SimpleRNN, Dense, Dropout
8
9
10 # -----
11 # 1. Parameters
12 # -----
13
14 vocab_size = 10000
15 max_length = 200
16
17
18 # -----
19 # 2. Load IMDB dataset
20 # -----
21
22 (X_train, y_train), (X_test, y_test) = imdb.load_data(
23     num_words=vocab_size
24 )
25
26 print("Training samples:", len(X_train))
27 print("Testing samples:", len(X_test))
28
29
30 # -----
31 # 3. Pad sequences
32 # -----
33
34 X_train = pad_sequences(
35     X_train,
36     maxlen=max_length,
37     padding='post',
38     truncating='post'
39 )
40
41 X_test = pad_sequences(
42     X_test,
43     maxlen=max_length,
44     padding='post',
45     truncating='post'
46 )
47
48 print("X_train shape:", X_train.shape)
49 print("X_test shape:", X_test.shape)
50
51
52 # -----
53 # 4. Build RNN model
54 # -----
55
56 model = Sequential([
57     Embedding(
58         input_dim=vocab_size,
59         output_dim=128,
60         input_length=max_length
61     ),
62     SimpleRNN(64),
63     Dropout(0.5),
64     Dense(1, activation='sigmoid')
65 ])
66
67
68 # -----
69 # 5. Compile model
70 # -----
71
72 model.compile(

```

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76     optimizer='adam',
77     loss='binary_crossentropy',
78     metrics=['accuracy']
79 )
80
81 model.summary()
82
83
84 # -----
85 # 6. Train model
86 # -----
87
88 history = model.fit(
89     X_train,
90     y_train,
91     epochs=5,
92     batch_size=64,
93     validation_split=0.2
94 )
95
96
97 # -----
98 # 7. Evaluate model
99 # -----
100
101 test_loss, test_accuracy = model.evaluate(
102     X_test,
103     y_test,
104     verbose=1
105 )
106
107 print("\nTest Loss:", test_loss)
108 print("Test Accuracy:", test_accuracy)
109
110
111 # -----
112 # 8. Plot results
113 # -----
114
115 plt.figure(figsize=(12, 5))
116
117 plt.subplot(1, 2, 1)
118 plt.plot(history.history['accuracy'], label='Training Accuracy')
119 plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
120 plt.xlabel('Epoch')
121 plt.ylabel('Accuracy')
122 plt.title('Accuracy')
123 plt.legend()
124
125 plt.subplot(1, 2, 2)
126 plt.plot(history.history['loss'], label='Training Loss')
127 plt.plot(history.history['val_loss'], label='Validation Loss')
128 plt.xlabel('Epoch')
129 plt.ylabel('Loss')
130 plt.title('Loss')
131 plt.legend()
132
133 plt.tight_layout()
134 plt.show()
135
136
137 # -----
138 # 9. Predict new review
139 # -----
140
141 word_index = imdb.get_word_index()
142
143
144 def encode_review(text):
145
146     words = text.lower().split()
147     encoded = []
148
149     for word in words:
150
151         index = word_index.get(word, 2) + 3
152
153         if index >= vocab_size:

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```
154         index = 2
155
156         encoded.append(index)
157
158     return pad_sequences(
159         [encoded],
160         maxlen=max_length,
161         padding='post',
162         truncating='post'
163     )
164
165
166 review = """
167 This movie was absolutely fantastic.
168 The acting was excellent and the story was very entertaining.
169 """
170
171 encoded_review = encode_review(review)
172
173 prediction = model.predict(encoded_review)[0][0]
174
175 print("\nReview:", review)
176 print("Sentiment Score:", prediction)
177
178 if prediction >= 0.5:
179     print("Prediction: POSITIVE")
180 else:
181     print("Prediction: NEGATIVE")
182
```

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz>
17464789/17464789 — 0s 0us/step
Training samples: 25000
Testing samples: 25000
X_train shape: (25000, 200)
X_test shape: (25000, 200)
/usr/local/lib/python3.13/dist-packages/keras/src/layers/core/embedding.py:100: UserWarning: Argument `input\_length` is deprecated
warnings.warn(
Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	?	0 (unbuilt)
simple_rnn (SimpleRNN)	?	0 (unbuilt)
dropout (Dropout)	?	0
dense (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)
Trainable params: 0 (0.00 B)
Non-trainable params: 0 (0.00 B)

Epoch 1/5

313/313 — 20s 59ms/step - accuracy: 0.4985 - loss: 0.7043 - val_accuracy: 0.4962 - val_loss: 0.6947

Epoch 2/5

313/313 — 22s 64ms/step - accuracy: 0.5811 - loss: 0.6676 - val_accuracy: 0.5040 - val_loss: 0.7008

Epoch 3/5

313/313 — 20s 65ms/step - accuracy: 0.6826 - loss: 0.5443 - val_accuracy: 0.4998 - val_loss: 0.7743

Epoch 4/5

313/313 — 18s 56ms/step - accuracy: 0.7442 - loss: 0.4269 - val_accuracy: 0.4948 - val_loss: 0.8775

Epoch 5/5

313/313 — 18s 57ms/step - accuracy: 0.7685 - loss: 0.3793 - val_accuracy: 0.5082 - val_loss: 0.9096

782/782 — 7s 10ms/step - accuracy: 0.5075 - loss: 0.9173

Test Loss: 0.9173304438591003

Test Accuracy: 0.507520020080872

